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PROJECT TECHNICAL REPORT

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PROJECT TITLE: SMART RENT — Rental Property Management System

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List of Abbreviations

- HTML — HyperText Markup Language
- CSS — Cascading Style Sheets
- JS — JavaScript
- UI — User Interface
- UX — User Experience
- ERD — Entity-Relationship Diagram
- DOM — Document Object Model
- DBMS — Database Management System
- CRUD — Create, Read, Update, Delete
- MVP — Minimum Viable Product

Chapter 1. Introduction

1.1. Historical Background of the Case Study

Rental property management in many residential and commercial settings, including within Rwanda, has traditionally relied on manual, paper-based processes or informal communication channels such as phone calls and physical visits. Landlords and property managers typically track tenant information, rent due dates, and maintenance requests using notebooks, spreadsheets, or memory, which introduces delays, miscommunication, and lost records. As property portfolios grow and tenant expectations shift toward faster, more transparent communication, the need for a simple digital platform that centralizes tenant registration, rent tracking, and maintenance requests has become increasingly evident.

Smart Rent was conceived as a response to this gap: a lightweight, web-based rental management tool that allows tenants, landlords, and property managers to register and interact with rental information through a single, accessible interface, without requiring complex software or specialized training.

1.2. Problem Statement

Landlords and property managers currently lack an easy, low-cost digital channel through which tenants can register their details and submit property-related requests. Existing solutions are either too complex and expensive for small-scale landlords, or too informal (phone calls, paper forms) to guarantee accurate record-keeping. This results in:

- Delayed or lost tenant registration information.
- Poor visibility into rent payment status and due dates.
- Slow handling of maintenance and property-related requests.
- Disorganized communication between tenants, landlords, and property managers.

Smart Rent addresses this by providing a simple, web-accessible registration and information system as the foundation for a broader rental management platform.

1.3. Objectives of the Project

General Objective

To design and develop a simple, user-friendly web-based platform (Smart Rent) that allows tenants, landlords, and property managers to register and manage core rental information in one place.

Specific Objectives

- To design a clear and responsive home page presenting the purpose and features of the Smart Rent system.
- To design and implement a registration form that captures tenant, landlord, and property manager details, including user type and property address.
- To implement client-side form validation and interaction using JavaScript.
- To design a conceptual data model (ERD) that will support future backend development for rent tracking and maintenance requests.
- To document the system analysis, design, and implementation process in line with the Web Design module requirements.

1.4. Expected Outcomes

- A functional, responsive two-page website (Home and Registration) built with HTML5, CSS3, and JavaScript.
- A working registration form with input validation and confirmation feedback for users.

- A clear visual identity (branding, layout, colour scheme) that reflects the purpose of a rental management platform.
- A documented system design (architecture and ERD) that provides a roadmap for extending the project with a backend, database, and authentication in future iterations.

1.5. Project Rationale (Importance to the Community)

Smart Rent contributes to the community by simplifying an everyday but often disorganized process: rental registration and management. For landlords and property managers, the system reduces the administrative burden of manually tracking tenant information. For tenants, it offers a straightforward way to register interest in a property and communicate requests without needing to visit an office in person. As a student project, it also demonstrates practical application of front-end web design principles — semantic HTML structure, responsive CSS layout, and interactive JavaScript — to a real-world problem relevant to the local housing and property management context.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1 Intended Users of the Project

- Tenants — individuals seeking to rent a property, who register their details and can submit requests or messages.
- Landlords — property owners who need visibility into tenant registrations, rent status, and maintenance requests.
- Property Managers — intermediaries who manage one or more properties on behalf of landlords and coordinate with tenants.

2.1.2 Intended System Partners

- Real estate and property management companies that could adopt the platform to digitize their tenant onboarding process.
- Independent landlords managing a small number of rental units who need a low-cost registration and communication tool.
- Future integration partners such as mobile money or banking APIs for rent payment (planned for later phases, not part of the current implementation).

2.2. System Design

1. High-Level Architecture

The current implementation is a static front-end web application consisting of two linked HTML pages (homepage.html and formpage.html), a shared stylesheet (style.css), and a client-side script (script.js) that handles form interaction. The architecture below also shows the planned backend and database layer required to make the system fully functional (persisting registrations, tracking rent payments, and managing maintenance requests), which is scoped as future work beyond the current front-end deliverable.

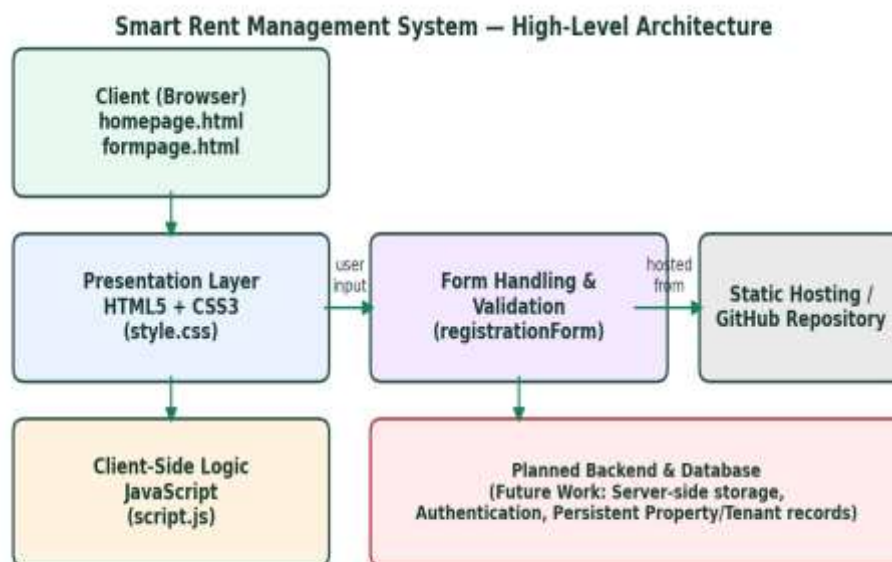


Figure 2.1: High-Level System Architecture

2. UI Design (Sketch of the UI Module) and Flowcharts

The user interface follows a simple two-screen flow: a Home page that introduces the system and its core features (rent tracking, maintenance requests, real-time updates) and directs users to register, and a Registration page containing the data-capture form. The navigation flow is:

- User lands on the Home page → reads feature highlights → clicks "Register Now".
- User is taken to the Registration page → fills in personal and property details → selects a user type (Tenant, Landlord, or Property Manager) → submits the form.
- The form is validated client-side; on successful submission, a confirmation message is displayed and the form resets.

Both pages share a consistent header with the site title and a navigation bar (Home / Register), giving the interface a unified look and feel across the two screens, as shown in the screenshots in Chapter 3.

3. Database Design with ERD and Relationships

Although the current version of Smart Rent operates entirely on the client side (no live database is connected yet), a conceptual data model was designed to guide future backend development. The proposed relational model centres on a USER entity linked to PROPERTY, REGISTRATION, RENT PAYMENT, and MAINTENANCE REQUEST entities, reflecting the fields already present in the registration form (full name, email, phone, user type, property address, and message) plus the additional entities needed to fully realise the objectives stated in Chapter 1.

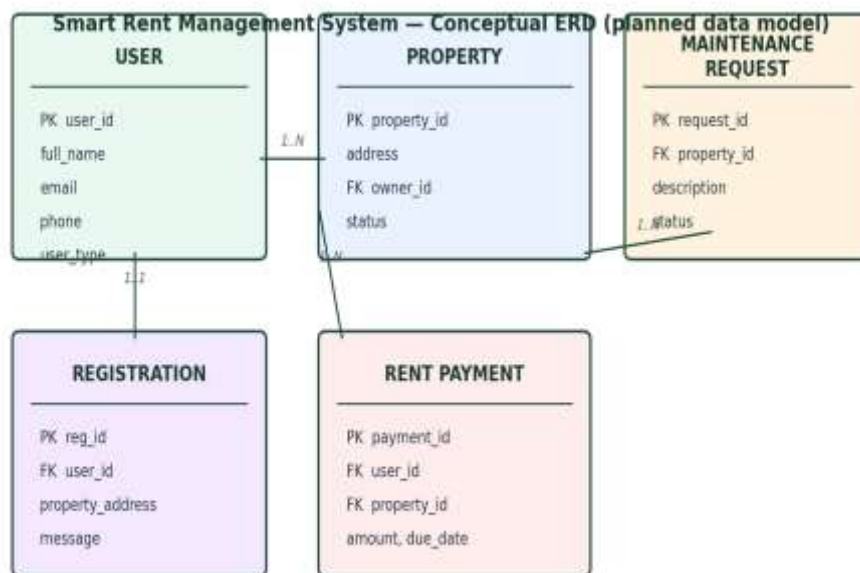


Figure 2.2: Conceptual Entity-Relationship Diagram (ERD)

Relationships: a USER (landlord) can own multiple PROPERTY records (1..N); each PROPERTY can have multiple MAINTENANCE REQUEST records (1..N); each USER submits one REGISTRATION record (1..1) capturing their initial sign-up details; and a USER can be associated with multiple RENT PAYMENT records over time (1..N), each linked to a specific property.

Chapter 3. Implementation

3.1. Introduction of the Section

This chapter presents the implemented screens of the Smart Rent system, built using HTML5 for structure, CSS3 for styling and responsive layout, and JavaScript for client-side interactivity. The system currently comprises two functional pages — the Home page and the Registration page — hosted in the project's GitHub repository. Each screenshot below is a live capture of the deployed pages from the repository, accompanied by a discussion of its purpose, structure, and the design decisions behind it.

3.2. Home Page

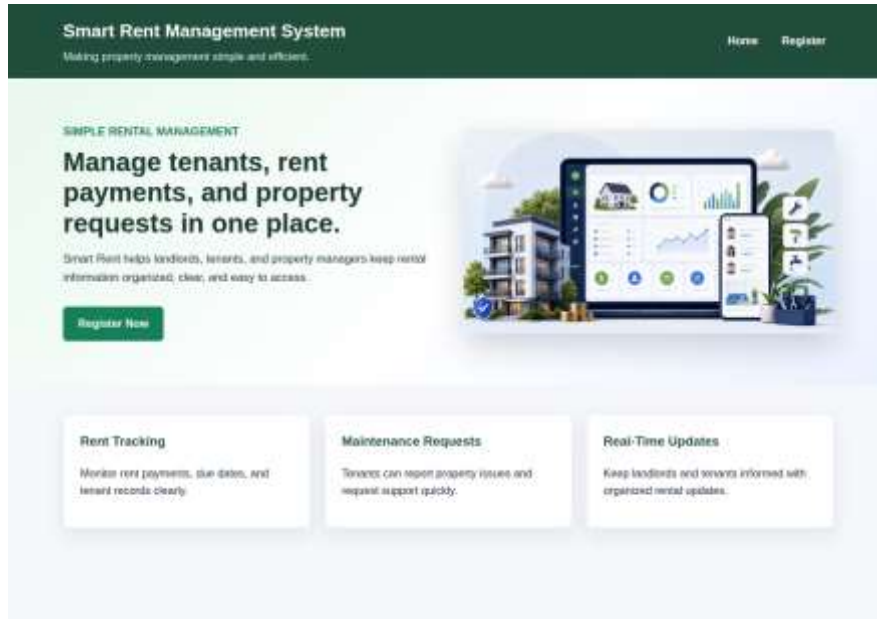


Figure 3.1: Home Page Screenshot (rendered from homepage.html in the project repository)

The Home page (homepage.html) serves as the entry point of the system. It uses a fixed header containing the system title, "Smart Rent Management System", a short tagline, and a navigation bar linking to the Home and Register pages. Below the header, a hero section introduces the value proposition ("Manage tenants, rent payments, and property requests in one place") alongside a call-to-action button, "Register Now", and a supporting illustration image.

Beneath the hero section, a three-column feature grid highlights the system's core planned capabilities: Rent Tracking, Maintenance Requests, and Real-Time Updates. This section was designed to communicate the full scope of Smart Rent's intended functionality to a first-time visitor, even though only the registration flow is currently interactive. The layout uses CSS Grid (feature-grid) for the cards and a two-column grid (hero-content) for the hero section, both of which collapse into a single column on smaller screens via a media query, ensuring the page remains usable on mobile devices.

Design choices such as the green-and-white colour palette (#16805A accent on a light background) were selected to evoke trust, stability, and a clean, professional feel appropriate for a property management brand. Rounded buttons and card shadows were used throughout to create visual depth without overwhelming the page.

3.3. Registration Page

Figure 3.2: Registration Page Screenshot (rendered from `formpage.html` in the project repository)

The Registration page (`formpage.html`) contains the system's core interactive element: the registration form (`id="registrationForm"`). It reuses the same header and navigation structure as the Home page for consistency, and introduces the form with a short explanatory paragraph clarifying that the page is shared by tenants, landlords, and property managers.

The form itself captures the following fields:

- Full Name (text, required) — identifies the registrant.
- Email (email, required) — used for confirmation and future communication.
- Phone Number (tel, required) — an alternative contact channel.
- User Type (dropdown, required) — Tenant, Landlord, or Property Manager, used to route the registration appropriately in a future backend.
- Property Address (text, optional) — relevant to landlords/managers registering a property, or tenants specifying interest in a property.
- Message / Request (textarea, optional) — a free-text field for additional context or requests.

Client-side validation is enforced through HTML5's required attribute and appropriate input types (email, tel), which prevent incomplete or malformed submissions before any data leaves the browser. On submission, the JavaScript file `script.js` intercepts the default form action, prevents a page reload (`event.preventDefault()`), displays a confirmation message ("Thank you! Your Smart Rent registration has been submitted successfully.") inside the `#formMessage` element, and resets the form fields. This provides immediate feedback to the user without requiring a backend at this stage of the project.

This screen fulfils the primary objective of the current milestone: a working, validated data-capture interface that can later be connected to a server-side endpoint and database once the backend is developed.

Chapter 4. Conclusion and Recommendation

State of the Implemented System

The Smart Rent project has successfully delivered the front-end foundation of a rental management platform: a responsive Home page and a fully validated Registration page, styled consistently and linked through shared navigation. Of the objectives defined in Chapter 1, the UI design, responsive layout, and client-side registration flow objectives are fully achieved (approximately 100% for the front-end scope).

However, the broader vision of a complete rental management system — persistent storage of registrations, live rent tracking, maintenance request handling, and real-time updates — has not yet been implemented, since no backend or database is currently connected. Considering the full set of objectives implied by the system's feature list (rent tracking, maintenance requests, real-time updates), the project is estimated to be at roughly 40–50% completion, with the remaining work concentrated on backend development.

Recommendations

To the Lecturer, about Technicalities

The registration form currently references its script using a relative path (`../js/script.js`) that does not match the actual file location in the repository root (`script.js`). This is a minor but instructive bug worth flagging in feedback, as it silently breaks the confirmation-message feature if not corrected. Reviewers are encouraged to verify script paths when the project is opened directly from the folder structure rather than a local server.

To the School, about Time Constraints

Given the module's assessment timeline, more dedicated studio/lab time for backend integration (even a lightweight one, such as a simple Node.js/Express API with a file-based or SQLite database) would allow students to demonstrate the full CRUD lifecycle rather than a front-end-only submission.

To the Partners (Property Owners/Managers Targeting This App)

Landlords and property managers considering adoption should treat the current version as a proof-of-concept for the registration workflow only. A production rollout would require a secure backend, data validation on the server side (not just the client), and safeguards for storing personal contact information in compliance with data protection expectations.

To the User, about Training

Because the interface is intentionally simple (a single form with standard input types), minimal training is required. A short onboarding note explaining the difference between the three user types (Tenant, Landlord, Property Manager) would help first-time users select the correct option.

To the Beneficiaries (Tenants/Customers)

Tenants and customers should be informed that, at this stage, the confirmation message displayed after submission is a client-side acknowledgement only and does not yet guarantee that the request has been stored or forwarded to a landlord, since no backend currently persists the data. This should be clearly communicated once the system is piloted with real users, to manage expectations until the backend is completed.

Appendix

GitHub Repository (source code of the deployed project):

<https://github.com/Mukamisha13/smart-rent-project>

Repository access credentials (login/password), if required by the module for private access, should be shared separately and securely with the lecturer and are not included in this document for security reasons: [Provide access details separately as instructed by the module coordinator].

Repository contents at the time of this report:

- homepage.html — Home page markup
- formpage.html — Registration page markup
- style.css — Shared stylesheet for both pages
- script.js — Client-side form handling logic
- hero-smart-rent.png — Hero illustration used on the Home page